

Analysis, Conclusions, and Hypotheses of Explanation

Cleaned Working Draft

Formalized from dictation. No theory or concept has been removed; the material has been de-colloquialized, de-duplicated across the Analysis and Conclusions, and reorganized into themed points. Overlapping items are stated once, in full, and referenced rather than repeated. Constants and mangled terms restored to their standard forms.

Analysis

The experiment behaved exactly as the framework predicts. That the results lined up as cleanly as they did was not guaranteed in advance, but in retrospect it is unsurprising: the mathematics is already present in general relativity, and the framework simply reads it in the two-system, scale-free form.

1. Scale preference and the relative nature of relativity

The central observation is that general and special relativity, as ordinarily presented, carry a strong **scale preference**: a scale-preferential view of the universe centered on the human scale. We have confronted and largely accepted this problem for space, but we have never confronted it for time. Time continues to be treated as absolute — our time taken as the preferred clock, everything else measured relative to it — even though time is governed by a theory named *relativity*.

The deeper problem is that relativity is not generally understood to be relative. Physics consensus does not clearly assert the relative nature of relativity, even though some individuals understand it. This is the reason for the paper's title: relativity is always relative. Everything observable is relative; there is nothing one can observe that is not.

2. The fallacy of the negligible

A recurring error underlies the scale-preferential habit. We look at a quantity that falls off toward infinity, becoming very small, and judge that by our own frame it is too small to matter. This is a sound *practical* judgment — it is a reasonable call about relevance to our frame of reference — but it is not a *mathematical* result, and it does not make our frame preferential.

This fallacy surfaces in a family of interchangeable phrases, all amounting to the same move:

negligible · almost nothing · infinitesimal · practically nonexistent · flat spacetime · time stoppage · approaching time stoppage · time stops · “the domain shifts” · “diverges from meaningful results”

Many of these are used loosely. In particular, there are no genuine *divergences* here; the term is frequently not used in its proper sense.

3. Flat spacetime is asymptotic — there is always “flatter”

Near our own scale, presuming that we calculate time dilation against flat spacetime causes little trouble. But flat spacetime is never actually reached. Choose any flat spacetime one can construct, and there is an infinity of flatter spacetime beyond it. No matter how flat a spacetime appears, there is always flatter.

Crucially, the picture is the same at every scale of flatness. Standing at any such scale, an observer would see the surroundings as just as curved as they appear to us locally — identical. One never obtains a qualitatively different picture at that scale. This is how time works and how physics works: it is always the same at the observer’s scale. Even as an observer grows, it remains the same at their *current* scale — which correctly implies that it is *different* between their prior scale and their current one.

4. Scale as lived experience — the childhood analogy

We are among the rare things that actually get to occupy different scales over a lifetime. As children, adults seemed very large; as adults, children seem very small. That shift is a change of scale perspective. We begin with small, high voices that deepen; we become large, bulky beings quite different from the children we were. This alters both the spatial and the temporal experience of the whole body.

Yet a body’s cells do not experience time differently from another body’s cells: they are of roughly the same constituency, size, and density. A brain may scale larger or smaller, but the pace of time within its cells and nerves stays consistent — which is essential for temporal measurement. The body need not perceive what we calculate; it simply runs at a pace relevant to itself. On average, we do not metabolize as fast as a smaller organism, nor as slowly as a larger one (not invariably, but on average).

5. No system without mass and radius

The framework implies that **there can be no system without mass and radius** (at least some of each). Two conventional objects are drawn into question by this:

- the **photon**, and
- the **singularity**.

I hold no personal stake in either and no reluctance to challenge either if it does not hold up. The point for the analysis is that both imply an *unbounded scale* that other views of the cosmos do not necessarily support.

The same reasoning bears on the **interpretation of the CMB**. The CMB is sometimes offered — though rarely admitted outright — as evidence for a *limit* to scale. Such a limit is not congruent with an inertia-based theory, which requires that everything in a system possess some mass and radius. To posit anything with neither is itself a scale-preferential move.

6. General relativity as the governing factor

The framework indicates that general relativity is the real governing factor, and that it does so without an apparatus of many distinct field models. (The charge-based models have not been

examined here.) Notably, the Kerr and ordinary black holes are described equally well by the Schwarzschild metric expanded to three dimensions — which is the natural setting for an inertial model of this kind.

7. Redshift as inertial time dilation, not expansion

A clear expectation follows: substantial redshift when viewing distant objects that are **not themselves** deep gravitational wells. Such objects may *contain* deep wells without constituting one, and ordinary intuition says they should therefore not redshift. But relative to us they carry significant **inertial time dilation** — the broader package of relative time dilation — and so exhibit that slower pace of time. (It is, in the end, purely gravitational: general relativity already accounts for the broad, low-density black holes that sit at the centers of galaxies, up to and including the entire observable universe.)

How to measure it. Redshift then carries a clean, usable meaning: it reports the spatial distribution of mass. The measurement is two-dimensional — mass over mean radius for the circle of sky under view:

For the circular field the telescope subtends, compute $m/\bar{r}$ (mass over mean radius) of what lies within it. For a galactic cluster, measure the cluster; for the region *including* the cluster, include all of it. Because $m/\bar{r}$ yields only the average within that sphere, finer granularity is obtained with a smaller sample.

What this replaces. Instead of extracting an ambiguous “age of the universe” that must be repeatedly recalculated — from a model so diluted it arguably no longer carries meaning, and one that has abandoned the singularity, leaving little beyond an explosion and an expansion that itself requires no explanation — general relativity provides an adequate account of the *entirety* of the redshift. We can then read the spatial distribution of mass directly, and the reading is validatable. Speed of expansion and system age are, by contrast, not meaningful quantities in this account.

On expansion and falsifiability. There is heavy conviction behind expansion, but a few decisive facts would settle whether it is valid, and there is ample indication it is not — beginning with its predictive record, which has required many revisions of the theory’s meaning (not the mark of a solid theory). The strongest point is simply that the effect is *already predicted by general relativity*. The essential test is narrow:

- Does general relativity predict what we observe?
- Does it predict *more* or *less* than we observe?

It predicts *less* — leaving a genuine remainder (“extra stuff”) to be explained. Were it to predict *more* than we see, the framework itself would be in trouble, and that possibility is left open rather than assumed away. These are easy validations, achievable with well-characterized, nearby systems such as Andromeda.

8. Unification of general and special relativity through inertia

This is a genuine unification of general and special relativity — developed in the preceding papers, and further validated here:

- **General relativity** is carried by **rotational inertia** (moment of inertia).
- **Special relativity** is carried by **ordinary, linear inertia** (mass).
- General relativity is, in this reading, *born of the moment of inertia*.

The premise that yields this can be stated and tested directly:

Presume that the inertia of any object or system — about a chosen axis, or along a chosen linear dimension — **is** the pace of time for that whole system. Rationally follow the consequences.

Special and general relativity emerge necessarily from that premise, which makes it nearly inconceivable that the premise is false. The definition of inertia supports it. Inertia (linear or rotational) is the property of an object or system that translates an applied force into the resulting change in motion. Any two systems with the same moment of inertia, under the same applied torque or force, undergo the same temporal effect — regardless of how each attained that moment of inertia. In this sense the **moment of inertia is literally system time**. Taking inertia’s additive, conservational nature together with this gives a clean, logical model of causality.

None of this changes what we observe or how we perceive it. It is a different *frame of observation* — the **system frame**. We ordinarily adopt a system frame and presume it to be the universal frame; it is not. Whether or not it is, and whether one calls the effect “time dilation” or “time scale,” it is the same thing. Einstein’s word would be *indistinguishable* — but no citation is required, because this is general relativity. To attack this is to attack general relativity.

9. Vision and color as a redshift mechanism (hypothesis)

A striking consequence: human vision very likely uses a **redshift mechanism** to perceive color. If it does not, that would point to a serious problem with the theory — so it is a real test.

The geometric requirement, and why the eye meets it. For the mechanism to work, the sensing element must sit at a consistent position relative to the incoming light. The chromophores do: they are mounted to the retina, which is always perpendicular to the incoming light. Every chromophore therefore receives light at a consistent angle (each at its own angle, set by its position relative to the center).

The mechanism. The cone-shaped chromophore bends at its middle when struck by a particular wavelength. Bending in the right direction covers its photosensitive portion and halts the reaction; it then relaxes back to a straight structure, ready to be excited again when energy of that frequency arrives. We carry three copies of the same chromophore across three cone sizes. Each shields the light’s angle just enough to sample a slightly different range of **inertial density**; from the three cones together, the system triangulates where the incoming signal falls within that range.

What color actually reports. Light mixes, scatters, reflects, and refracts from a source of emission; when it falls in the visible range it triggers the three chromophores and we register a color. What the eye is really reading is **mass over radius about the axis of observation**. Taking a chromophore as it faces the plane perpendicular to the incoming light and computing its mass over mean radius — its shape and mass are both well enough understood to model as point masses in the correct positions — yields a value on the order of 10^{-43} (in the corresponding units).

Color $\leftrightarrow$ inertial density. Denser systems (more mass concentrated in a smaller region) have a smaller moment of inertia and therefore a faster pace of time. As rough magnitudes:

Appearance	Order of magnitude
white / yellow	$\sim 10^{-43}$
red	$\sim 10^{-43}$
blue	$\sim 10^{-44}$

Beyond these ranges one passes into infrared (redder) or ultraviolet (bluer). Smaller, denser systems trend toward smaller moment of inertia and thus blueshift (or less redshift); reds, oranges, and purples toward the other end. This appears to hold whether the colorant is a solid, liquid, or gas, a pigment or otherwise — the same mechanism is passed through.

Chemistry cross-check. Chemistry explains these colors by electron transitions, and for larger, sophisticated coloring molecules (chromophore-type molecules) that description matches the behavior. Gases are similar — but the **noble gases** have no good electron-based explanation and are regarded as an anomaly for electron-transition accounts. From the inertial-density view there is no anomaly: inertial density rises as inertia increases over the same volume, up to the DeGerlia Threshold $D_{\text{crit}} = 6.73295 \times 10^{26}$ kg/m. The radius at which electrons orbit the nucleus — especially in higher-order systems — bears materially on the moment of inertia, enough to shift the time dilation between observing and observed systems. Neon, for instance, becomes bluer when compressed and redder when the pressure is reduced. The same reasoning covers small-particle systems that sit in the UV and trickle into the visible — water, or the daytime blue sky (an illuminated system of objects that blueshift relative to the eye’s chromophore).

Predicting resonance from geometry. The decisive test would be predicting a compound’s UV/visual resonance from geometry alone, which appears feasible. Taking water — present as H_2O , H_3O^+ , and OH^- , with a few other transient forms to account for — one computes each form’s mass-to-radius ratio about each of its three axes to obtain a range of resonance per axis; weighting by proportion yields a representative peak for each three-dimensional orientation of the compound. Water viewed face-on presents a broad profile; viewed edge-on (down the O–H bond) it is flat — the same mass, a different view — and the picture grows more complex as the system grows larger. This opens applications: engineering compounds to meet a target absorption pattern — made *uniform* across viewing axes, or deliberately *non-uniform* so that a material reads as one color from one angle and a different color from another.

10. Assessment of the framework itself

The framework proved very useful and exhibited every characteristic expected of it; it is a compatible model. The salient finding concerns the **scale-dilation term**:

- Either scale dilation has been rendered entirely out of gravitational time dilation, or — more likely — it has been *interpreted away* in a human-centric manner until it carries no meaning. That kind of drift is how word-meanings shift over time, and it is not always illegitimate.
- As hypothesized, the normalization that drives the value into $[0, 1]$ is flawed, or is at least routinely interpreted in a flawed way. It carries a factor that, if presumed negligible, is simply lost — and it should never be presumed negligible with respect to the functioning of whatever one is working on.

The framework does make the time-dilation changes clear and intuitive. In the classical worked example the scale factor came out to 2, exactly as expected — a purely spatial effect, with time dilation negligible (a value like $2.000\dots$ with roughly twenty zeros before any nonzero digits).

The real takeaway concerns **DTD versus GTD**. DTD ($\sqrt{k}$) is a faithful representation of time dilation that GTD does not necessarily capture. Removing the “1–” of the GTD form ($\sqrt{1-k}$) leaves a remarkably clean and complete framework; GTD, by comparison, gives only a shadow of it. The two carry the *same information* — but one cannot fully map back to the original systems from either quantity alone without the actual real-world parameters of at least one system.

Conclusions

Several of these conclusions restate results asserted earlier — in the analysis above, and repeatedly across this framework and the prior papers. That restatement is deliberate, not redundant: a conclusion that confirms a result already asserted is *confirmation*, and the confirmation is itself the result. Each point below is drawn here as a firm conclusion, with the supporting discussion in the corresponding Analysis section.

1. **Relativity is relative — there is no privileged frame or scale.** Everything observable is relative; there is nothing one can observe that is not. Time is no exception, and it is the one place where the habit of treating a quantity as absolute has gone unchallenged (§1). Any observer’s account of how time flows looks different from any other’s, yet no two such accounts describe two *different* systems — they describe the **same** system at the **same** moments, observed differently. This is what it means to explain the relative nature of relativity.
2. **Flat spacetime is asymptotic; it is never reached.** For any spacetime one can construct there is an infinity of flatter, and at every scale of flatness the surroundings appear just as curved as ours do to us locally — the picture is identical at every observer’s scale (§3). That flat spacetime is asymptotic is not in dispute; what follows from taking it seriously is.
3. **Human-centric negligibility corrupts results outside the core regime.** Treating flat spacetime and negligibility from the human scale distorts how gravitation is understood (§2, §3), and it makes results *numerically wrong* once one drifts out of the core regime in which humans are presumed to be the measure of time. Adopting the human scale as the basis of measurement is acceptable *as a convention* — but it is only that.
4. **General and special relativity unify through inertia.** General relativity is carried by rotational inertia and special relativity by linear inertia; the moment of inertia is, in this reading, literally system time (§8). Two systems with equal moment of inertia under equal applied force undergo the same temporal effect, however each attained it. Because this follows directly from general relativity, to contest it is to contest general relativity.
5. **Every system has mass and radius; scale is unbounded.** The framework admits no system without some mass and some radius, and therefore no limit to scale (§5). This draws into question the photon, the singularity, and any reading of the CMB as evidence for a limit to scale.
6. **Two systems’ pace of time can be compared directly in DTD, straight from general relativity.** Confirmed here in the recently identified space–time–equivalence (STE)

form. For two systems compared about their axes of observation:

$$\text{DTD} = \sqrt{\frac{m_1 r_2}{m_2 r_1}} \quad (\text{axis of observation})$$

The DTD comparison is reached in a single step. The GTD comparison is not: one must compute each system’s value and then take the ratio. Though reasonably controversial, the DTD relation is a direct consequence of general relativity, not an addition to it.

Hypotheses of Explanation

These are items whose explanation I understand and hold to be right. I do not label them *theory* beyond what is warranted — but that restraint is the strength of a hypothesis of explanation, not a retreat from the result. That each of these was *predicted* by the framework already makes the case strong; what remains genuinely open is only the *meaning* one attaches to some of them. I have not exhaustively demonstrated, numerically and on paper, that every one is consistent in every case with the predictions of general and special relativity — but nothing here requires new observations. No new tests need be run; the observations would not change. The only open question is whether it **adds up** under this understanding — and it does.

The explanations put forward, each following from the framework and each already consistent with what we observe:

1. **Redshift is inertial time dilation, not cosmic expansion (§7).** Distant systems that are not themselves deep gravitational wells still carry significant inertial time dilation relative to us and therefore redshift. General relativity accounts for the entirety of the effect; the redshift then reads directly as the spatial distribution of mass ($m/\bar{r}$ over the field of view), a clean and validatable quantity — where “age of the universe” and “speed of expansion” are not.
2. **Human color vision is a redshift mechanism (§9).** The eye reads mass over radius about the axis of observation. The chromophores’ fixed perpendicular mounting to the retina, the cone-size triad, and the inertial-density-to-color mapping all follow from this, and it is a real, falsifiable test of the framework.
3. **A compound’s optical resonance is predictable from its geometry (§9).** Mass-to-radius ratios about a molecule’s three axes give its resonance ranges and peaks — including the noble-gas behavior (e.g. neon bluer under compression, redder under reduced pressure) that electron-transition accounts treat as anomalous.

I have every right to draw these explanatory conclusions, and this is the proper place to state them plainly. That is the beauty of a hypothesis of explanation: it is flexible. It sets out the account without the overreach of asserting settled theory, and it stays open to test rather than pretending to be closed. This is precisely the discipline the expansion account abandoned — had cosmic expansion been *held as a hypothesis* rather than assumed and asserted as established, it would not now look as forced as it does.

(By contrast, the unification through inertia (§8) and the two numbered Conclusions are held to a firmer standard than “hypothesis” — they follow directly from general relativity and are not offered tentatively.)